

# ZeroBT – Next- Generation Enterprise AI Customer Support

**Development Team:** Sanjog Raina · Pranav Singhal · Aditya Choudhari · Nateshi Babbilwar · Shivangi Kanojia · Aarti Parmar · Pratima Chavan





# The Problem & The Solution

## The Challenge

- Legacy chatbots hallucinate and lack escalation
- Budget/staffing constraints limit support scaling

## The ZeroBT Solution

- **100% Grounded Answers:** with citations
- **Real-Time Frustration Tracking**
- **Consent-Based Escalation**

# Technical Architecture & Core Stack



## Frontend UI

Simple, responsive chat interface for users



## Query Engine

Understands user questions and routes them intelligently



## Hybrid RAG Pipeline

Finds the most relevant information from documents



## Re-Ranking

Selects the best answers from the initial results



## Sentiment Engine

Detects frustration and triggers support when needed



## Voice & Multimodal

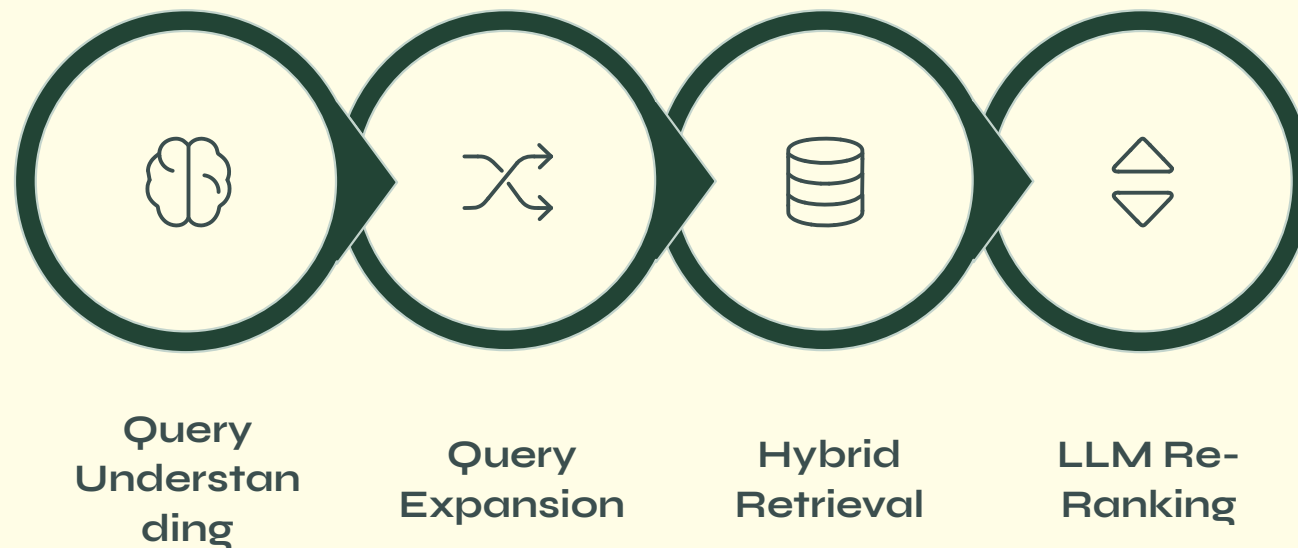
Supports text inputs and audio responses with varied speech speeds



## Escalation & Storage

Saves conversations and sends support alerts

# Real-Time NLP & Hybrid RAG Pipeline



## Query Understanding

- **Intent Classification:** Refunds, Shipping, Technical, Legal, Complaints
- **Entity Extraction:** SKUs, order numbers, policy names, locations

## Query Expansion

- **Search Variations:** Generates 10 variations per query
- **Recall Boost:** Improves coverage across phrasing differences

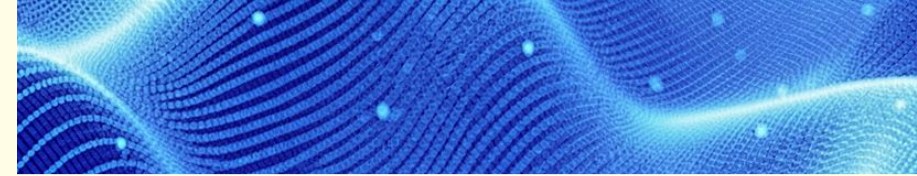
## Hybrid Retrieval

- **Dense Search:** Semantic matching with text embeddings
- **Sparse Search:** BM25 keyword matching for exact terms

## LLM Re-Ranking

- **Rank Fusion:** Merges candidate results into one shortlist
- **Precision Filter:** Promotes the most relevant passages

# Psychological Frustration Tracking & Escalation



## Real-Time Sentiment Analysis

- **Dynamic Frustration Score (0–100)** with color indicators: **Green** = Calm, **Orange** = Frustrated, **Red** = Severe
- **Risk Signal Detection:**  
demand\_human, threat,  
legal\_risk

## 8-Level Escalation Hierarchy

01

---

**Tier 1 & 2 Support**

02

---

**Tier 3 Technical → Tier 4 Policy**

03

---

**Senior Agent → Support Manager**

04

---

**Business Director → Founder**

# Key Metrics & Business Impact

**75%**

## **FRT Reduction**

Dramatic drop in First Response Time  
across all support channels

**95%+**

## **Grounded Accuracy**

Zero hallucination with strict inline citation  
of every fact

**40%**

## **Fewer Escalations**

Reduction in unnecessary Tier-1 human  
escalations